

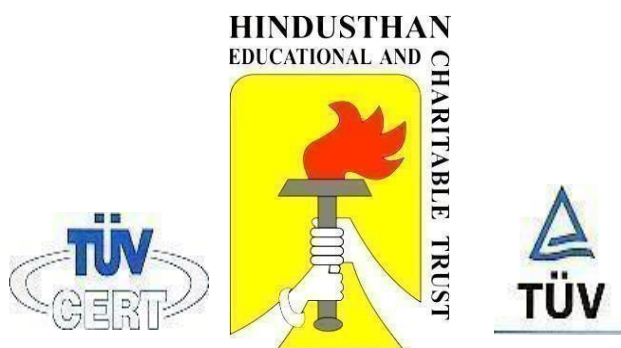
HINDUSTHAN COLLEGE OF ARTS & SCIENCE (AUTONOMOUS)

An Autonomous Institution - Affiliated to Bharathiar University

(ISO 9001-2001 Certified Institution)

Behind Nava India, Coimbatore – 641028.

DEPARTMENT OF COMPUTER APPLICATIONS (PG)



MASTER OF COMPUTER APPLICATIONS

PRACTICAL RECORD

25MCP21TP – PRACTICAL: BIG DATA ANALYTICS

NAME : _____

REGISTER NO : _____

CLASS : _____

SEMESTER : _____

YEAR : _____

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25MCP21TP – PRACTICAL: BIG DATA ANALYTICS

CERTIFICATE

Certified that this is a bonafide record of **BIG DATA ANALYTICS
(25MCP21TP)** done by _____

Register No: _____ during the academic year 2026-2027.

STAFF IN - CHARGE

HEAD OF THE DEPARTMENT

Submitted for the Bharathiar University Practical Examination held on _____
at Hindusthan College of Arts & Science, Coimbatore - 28.

INTERNAL EXAMINER

EXTERNAL EXAMINER

Date:

Place: Coimbatore

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EX.NO: 01

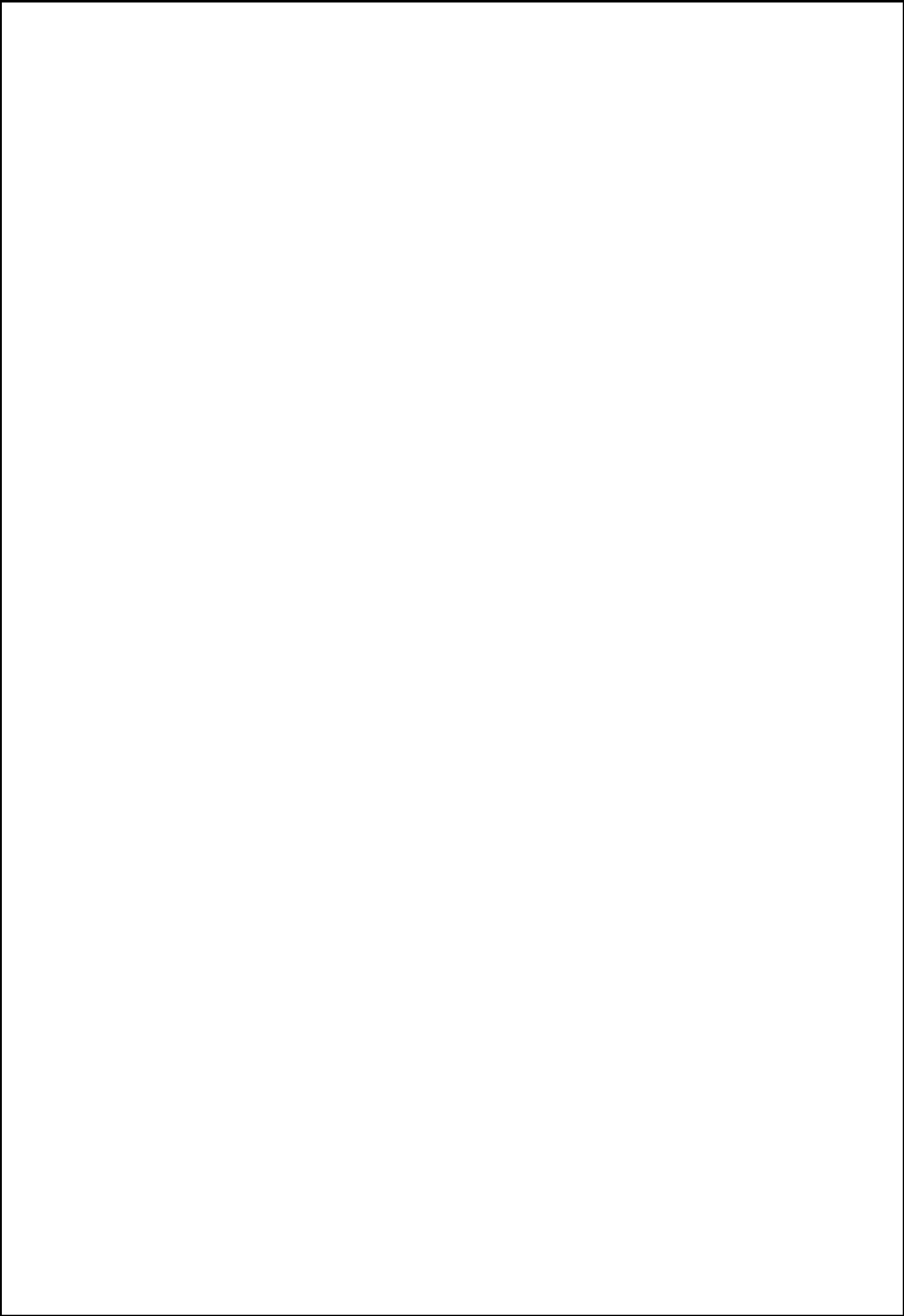
DATE:

**HADOOP THREE OPERATING
MODES**

PAGE NO:

AIM:

ALGORITHM:



SOURCE CODE:

STANDALONE MODE:

```
mkdir C:\hadooptest\input
```

```
echo Hello Hadoop World > C:\hadooptest\input\sample.txt
```

```
echo Hadoop is a Big Data framework >> C:\hadooptest\input\sample.txt
```

```
hadoop jar C:\hadoop\share\hadoop\mapreduce\hadoop-mapreduce-examples-3.3.6.jar ^  
grep C:\hadooptest\input C:\hadooptest\output "H[a-z]+"
```

```
type C:\hadooptest\output\part-r-00000
```

PSEUDO-DISTRIBUTED MODE:

```
start-dfs.cmd
```

```
start-yarn.cmd
```

```
jps
```

```
hdfs dfs -mkdir /user
```

```
hdfs dfs -mkdir /user/student
```

```
hdfs dfs -ls /
```

FULLY DISTRIBUTED MODE:

```
start-all.cmd
```

```
192.168.1.101
```

```
192.168.1.102
```

```
<configuration>
```

```
  <property>
```

```
    <name>fs.defaultFS</name>
```

```
    <value>hdfs://192.168.1.100:9000</value>
```

```
  </property>
```

```
</configuration>
```

OUTPUT:

STANDALONE MODE:

```
2  Hadoop
1  Hello
```

PSEUDO-DISTRIBUTED MODE:

jps Output:

```
NameNode
DataNode
ResourceManager
NodeManager
```

HDFS Output:

```
Found 1 items
drwxr-xr-x  - student supergroup  0  /user
```

FULLY DISTRIBUTED MODE:

```
Master Machine
192.168.1.100
|
|-- NameNode
|-- ResourceManager
|
|-- Worker 1
|   192.168.1.101
|   |-- DataNode
|   |-- NodeManager
|
|-- Worker 2
    192.168.1.102
    |-- DataNode
    |-- NodeManager
```

RESULT:

EX.NO: 02

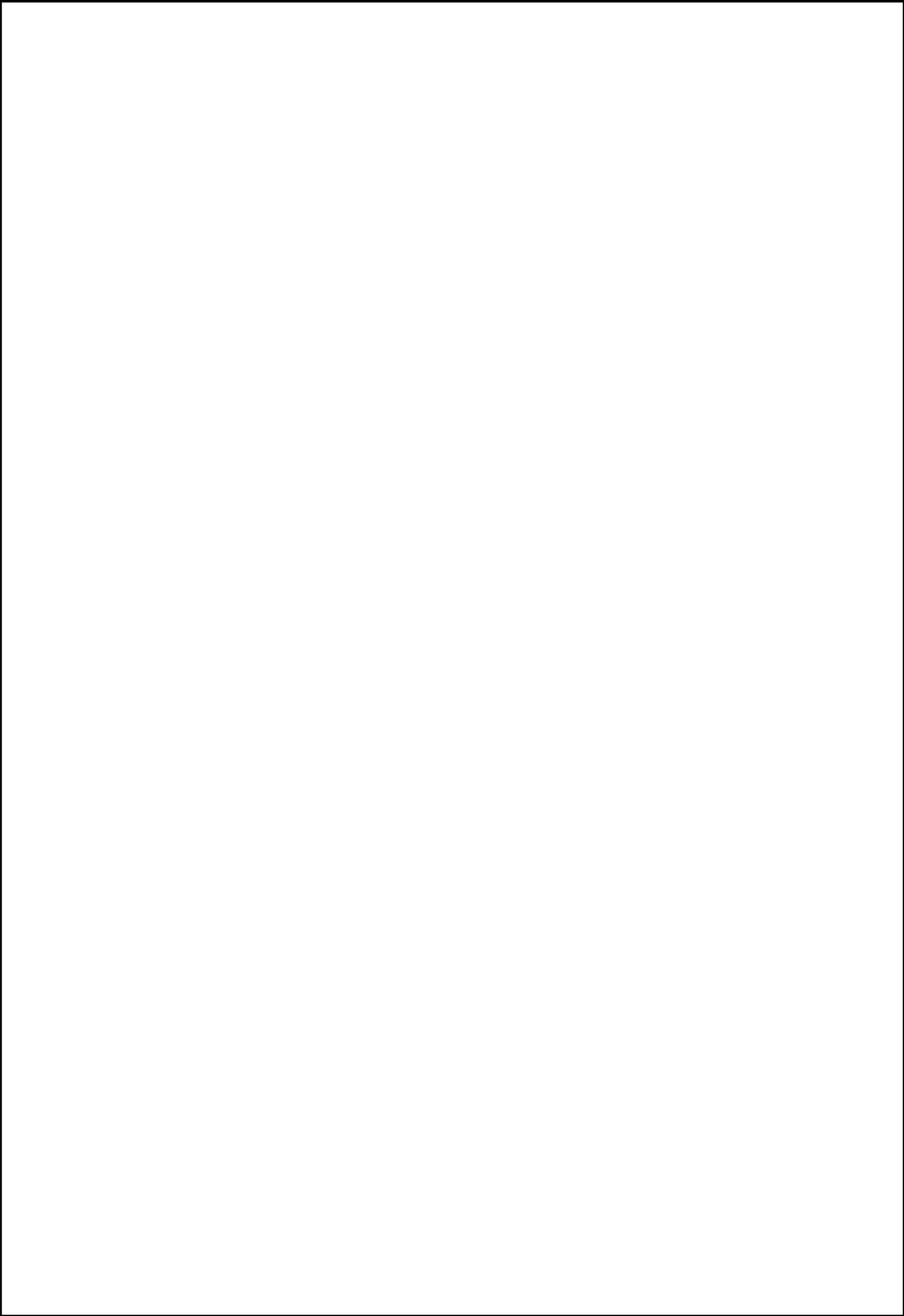
DATE:

HDFS FILE MANAGEMENT

PAGE NO:

AIM:

ALGORITHM:



SOURCE CODE:

ADDING FILES AND DIRECTORIES:

```
hdfs dfs -mkdir /user
hdfs dfs -mkdir /user/student
hdfs dfs -mkdir /user/student/data
mkdir C:\mydata
echo Name,Age,City > C:\mydata\students.txt
echo Ravi,21,Chennai >> C:\mydata\students.txt
echo Priya,22,Mumbai >> C:\mydata\students.txt
echo Karthik,20,Delhi >> C:\mydata\students.txt
hdfs dfs -put C:\mydata\students.txt /user/student/data/
hdfs dfs -ls /user/student/data/
hdfs dfs -du -h /user/student/
```

RETRIVING FILES FROM HDFS:

```
hdfs dfs -cat /user/student/data/students.txt
hdfs dfs -get /user/student/data/students.txt C:\mydata\downloaded.txt
type C:\mydata\downloaded.txt
hdfs dfs -cp /user/student/data/students.txt /user/student/backup.txt
hdfs dfs -ls -R /user/student/
hdfs dfs -df -h /
```

DELETING FILES:

```
hdfs dfs -rm /user/student/backup.txt
hdfs dfs -rm -r /user/student/data
hdfs dfs -rm -r -skipTrash /user/student/data
hdfs dfs -ls /user/student
```

OUTPUT:

ADDING FILES AND DIRECTORIES:

Directory created successfully

/user

/user/student

/user/student/data

Name, Age, City

Ravi, 21, Chennai

Priya, 22, Mumbai

Karthik, 20, Delhi

Found 1 items

-rw-r--r-- 1 user supergroup 59 /user/student/data/students.txt

59 /user/student/data

RETRIEVING FILES:

Name, Age, City

Ravi, 21, Chennai

Priya, 22, Mumbai

Karthik, 20, Delhi

Name, Age, City

Ravi, 21, Chennai

Priya, 22, Mumbai

Karthik, 20, Delhi

/user/student/data/students.txt

/user/student/backup.txt

/user/student

/user/student/backup.txt

/user/student/data

/user/student/data/students.txt

Configured Capacity: 105.44 GB

Present Capacity: 105.44 GB

DFS Remaining: 105.43 GB

DFS Used: 59 B

DFS Used%: 0%

DELETING FILES:

Deleted /user/student/backup.txt

Deleted /user/student/data

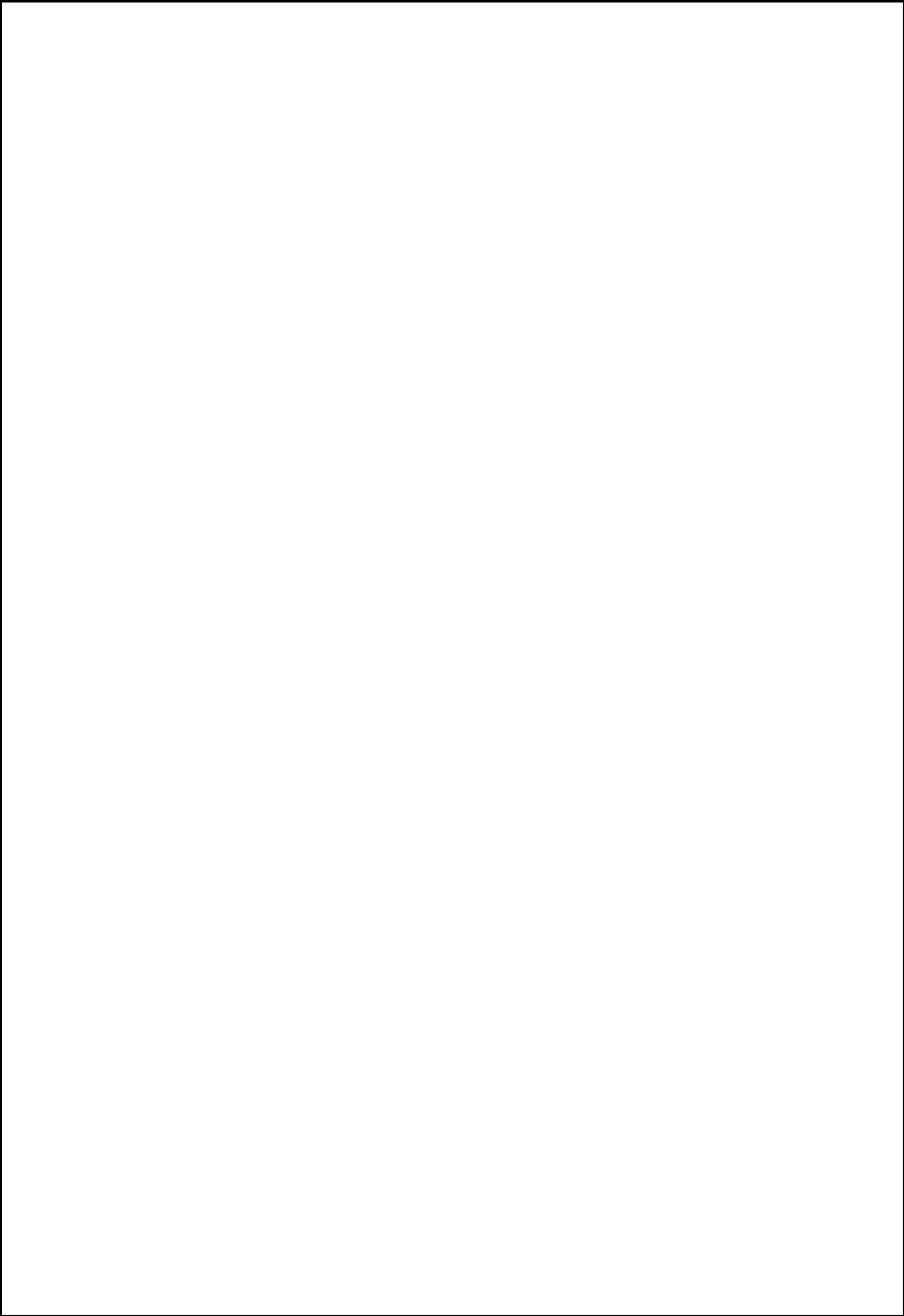
Found 0 items

RESULT:

EX.NO: 03	WORD COUNT (MAPREDUCE PARADIGM)	PAGE NO:
DATE:		

AIM:

ALGORITHM:



SOURCE CODE:

```
C:\Users\STUDENT>cd..
```

```
C:\Users>cd..
```

```
C:\>mkdir mapreduce
```

```
C:\>cd mapreduce
```

```
C:\mapreduce>mkdir wordcount
```

```
C:\mapreduce>cd wordcount
```

```
C:\mapreduce\wordcount>type wordcount.txt | python mapper.py
```

```
C:\mapreduce\wordcount>type wordcount.txt | python mapper.py | sort | python reducer.py
```

WORDCOUNT.TXT:

HINDUSTHAN COLLEGE OF ARTS AND SCIENCE
DEPARTMENT OF COMPUTER APPLICATIONS(PG)

MAPPER .PY :

```
import sys
```

```
for line in sys.stdin:
```

```
    line = line.strip()
```

```
    words = line.split()
```

```
    for word in words:
```

```
        print(f"{word}\t1")
```

REDUCER.PY:

```
import sys

current_word = None
current_count = 0

for line in sys.stdin:
    line = line.strip()
    word, count = line.split("\t", 1)
    count = int(count)

    if current_word == word:
        current_count += count
    else:
        if current_word is not None:
            print(f"{current_word}\t{current_count}")

        current_word = word
        current_count = count

if current_word is not None:
    print(f"{current_word}\t{current_count}")
```

OUTPUT:

MAPPER:

HINDUSTHAN 1
COLLEGE 1
OF 1
ARTS 1
AND 1
SCIENCE 1
DEPARTMENT 1
OF 1
COMPUTER 1
APPLICATIONS(PG) 1

REDUCER OUTPUT:

AND 1
APPLICATIONS(PG) 1
ARTS 1
COLLEGE 1
COMPUTER 1
DEPARTMENT 1
HINDUSTHAN 1
OF 2
SCIENCE 1

RESULT:

EX.NO: 04

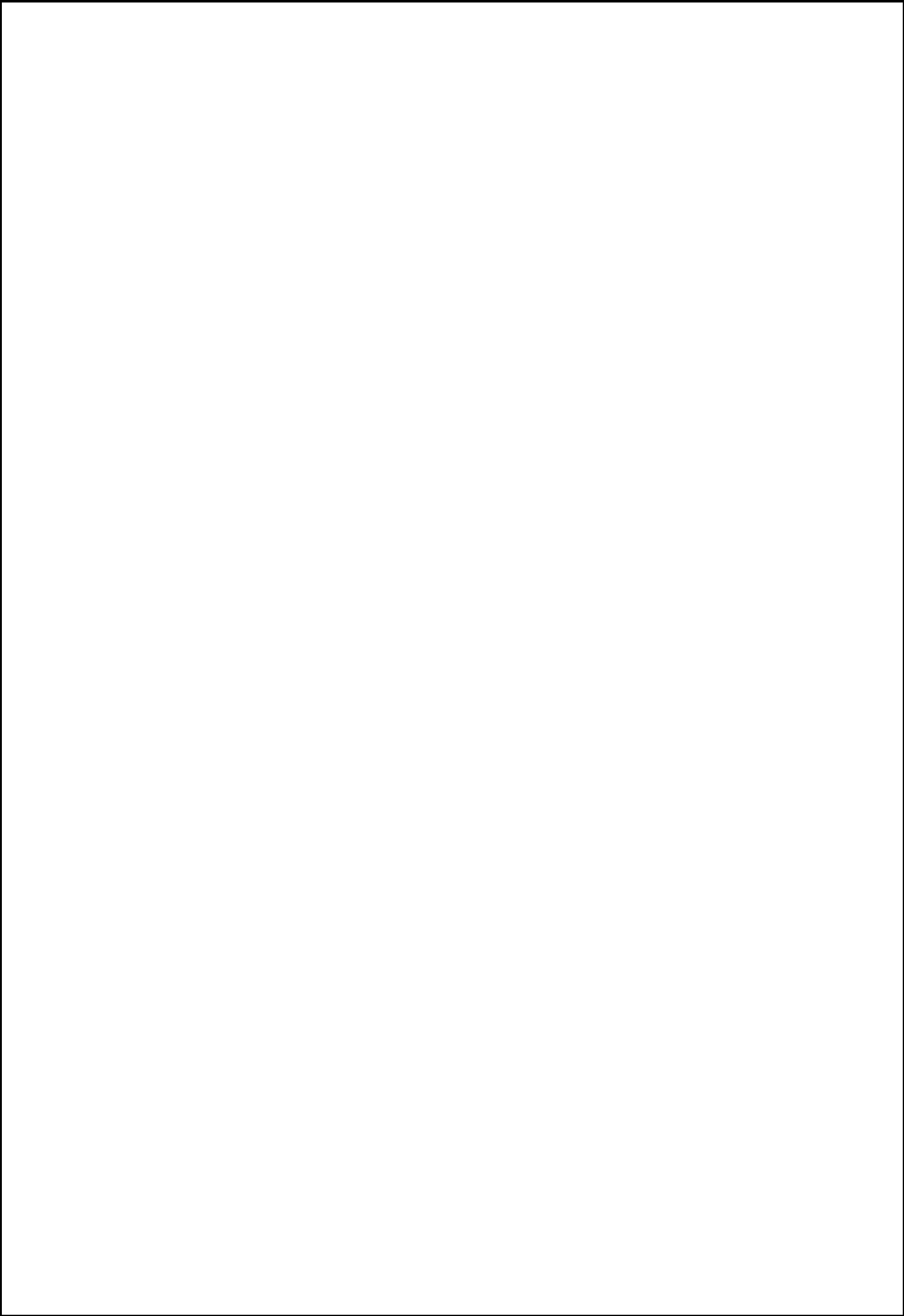
WEATHER DATA MINING

PAGE NO:

DATE:

AIM:

ALGORITHM:



SOURCE CODE:

```
C:\>cd hadoop
```

```
C:\hadoop>mkdir C:\mapreduce\weather_mining
```

```
C:\hadoop>cd C:\mapreduce\weather_mining
```

```
C:\mapreduce\weather_mining>notepad weather.csv
```

```
C:\mapreduce\weather_mining>type weather.csv
```

```
C:\mapreduce\weather_mining>notepad weather_mapper.py
```

```
C:\mapreduce\weather_mining>notepad weather_reducer.py
```

```
C:\mapreduce\weather_mining>type weather.csv | python weather_mapper.py | sort | python  
weather_reducer.py
```

weather.csv

```
Year ,City,    MaxTemp  
2019, Mumbai ,34.5  
2019, Delhi,   42.1  
2019, Chennai,38.3  
2020, Mumbai,35.2  
2020, Delhi,   40.0  
2021, Mumbai,36.1  
2021, Delhi,   43.5  
2022, Delhi,   41.7
```

Weather_mapper.py

```
import sys
```

```
first_line = True
```

```
for line in sys.stdin:
```

```
    if first_line:
```

```
        first_line = False
```

```
        continue
```

```
    line = line.strip()
```

```
if not line:
    continue

parts = line.split(',')

if len(parts) == 3:
    year = parts[0]
    temp = parts[2]

    print(year + '\t' + temp)
```

Weather_reducer.py

```
import sys

current_year = None

max_temp = float('-inf')

for line in sys.stdin:

    year, temp = line.strip().split('\t')

    temp = float(temp)

    if current_year == year:

        if temp > max_temp:

            max_temp = temp

    else:

        if current_year:

            print(current_year + '\t' + str(max_temp))

        current_year = year
```

```
max_temp = temp

if current_year:

    print(current_year + '\t' + str(max_temp))
```

OUTPUT :

MAPPER:

2019	42.1
2020	40.0
2021	43.5
2022	41.7

RESULT:

EX.NO: 05

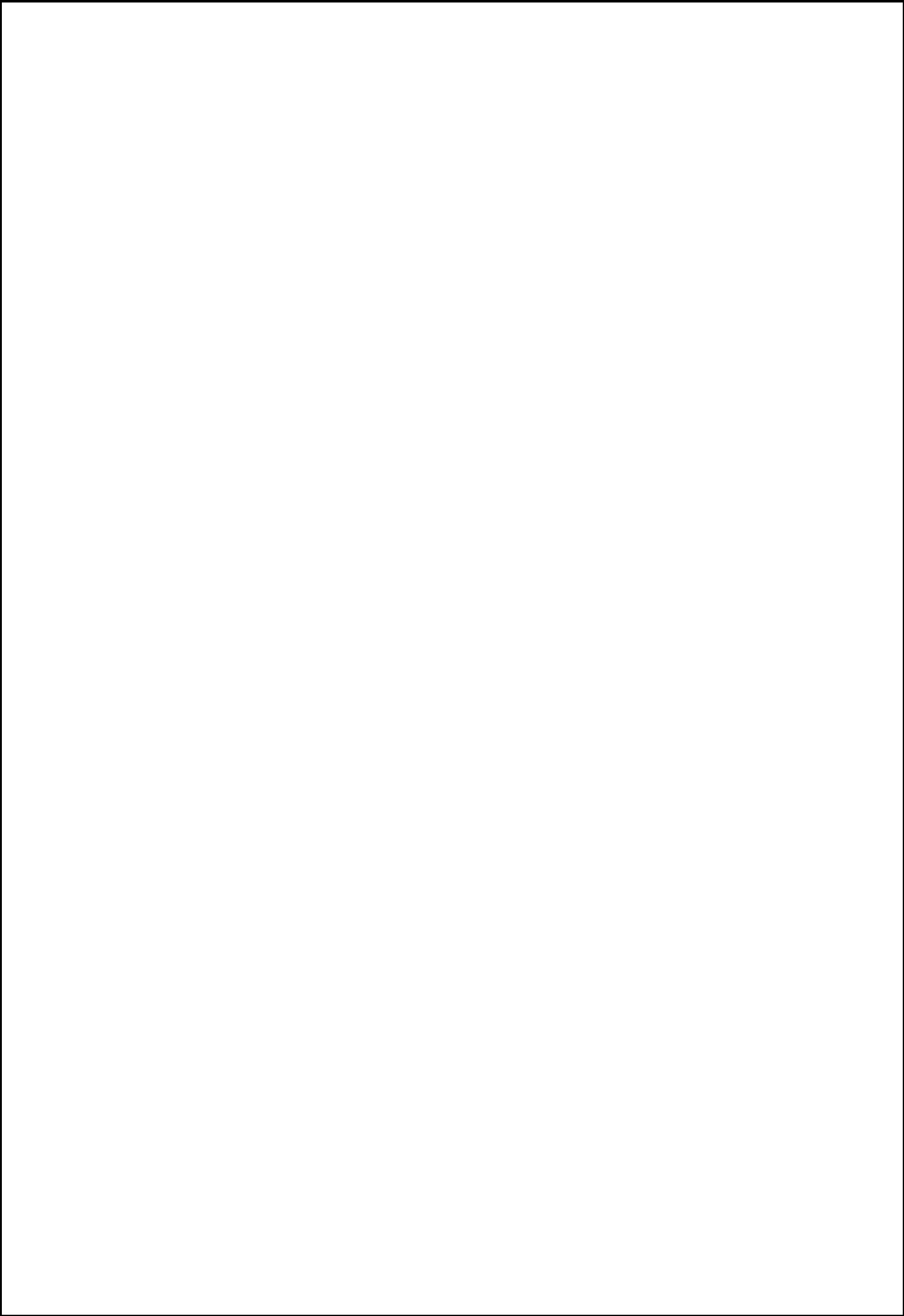
MATRIX MULTIPLICATION

PAGE NO:

DATE:

AIM:

ALGORITHM:



SOURCE CODE:

```
C:\Users\STUDENT>cd..\.
```

```
C:\>mkdir C:\mapreduce\matrix_mining
```

```
C:\>cd C:\mapreduce\matrix_mining
```

```
C:\mapreduce\matrix_mining>notepad matrix_input.txt
```

```
C:\mapreduce\matrix_mining>type matrix_input.txtmatrix_input.txt:
```

```
C:\mapreduce\matrix\matrix_input.txt
```

```
C:\mapreduce\matrix_mining>notepad matrix_mapper.py
```

```
C:\mapreduce\matrix_mining>notepad matrix_reducer.py
```

```
C:\mapreduce\matrix_mining>type matrix_input.txt | python matrix_mapper.py | sort |  
python matrix_reducer.py
```

Matrix_input.txt:

```
A,0,0,1
```

```
A,0,1,2
```

```
A,1,0,3
```

```
A,1,1,4
```

```
B,0,0,5
```

```
B,0,1,6
```

```
B,1,0,7
```

```
B,1,1,8
```

matrix_mapper.py:

```
import sys
```

```
m, n, p = 2, 2, 2
```

```
for line in sys.stdin:
```

```
    parts = line.strip().split(',')
```

```
    matrix = parts[0]
```

```
    i, j, val = int(parts[1]), int(parts[2]), float(parts[3])
```

```
if matrix == 'A':  
  
    for k in range(p):  
  
        print(str(i) + ',' + str(k) + '\tA,' + str(j) + ',' + str(val))  
  
else:  
  
    for i2 in range(m):  
  
        print(str(i2) + ',' + str(j) + '\tB,' + str(i) + ',' + str(val))
```

matrix_reducer.py:

```
import sys  
  
current_key = None  
  
a_vals = {}  
  
b_vals = {}  
  
def compute_cell(key, a, b):  
  
    result = 0  
  
    for k in a:  
  
        if k in b:  
  
            result += a[k] * b[k]  
  
    print(key + '\t' + str(result))  
  
for line in sys.stdin:  
  
    key, value = line.strip().split('\t')  
  
    parts = value.split(',')  
  
    matrix = parts[0]  
  
    idx = int(parts[1])
```

```
val = float(parts[2])
```

```
if current_key != key:
```

```
    if current_key:
```

```
        compute_cell(current_key, a_vals, b_vals)
```

```
    current_key = key
```

```
    a_vals, b_vals = {}, {}
```

```
if matrix == 'A':
```

```
    a_vals[idx] = val
```

```
else:
```

```
    b_vals[idx] = val
```

```
if current_key:
```

```
    compute_cell(current_key, a_vals, b_vals)
```

OUTPUT:

0,0	19.0
0,1	22.0
1,0	43.0
1,1	50.0

RESULT:

EX.NO: 06

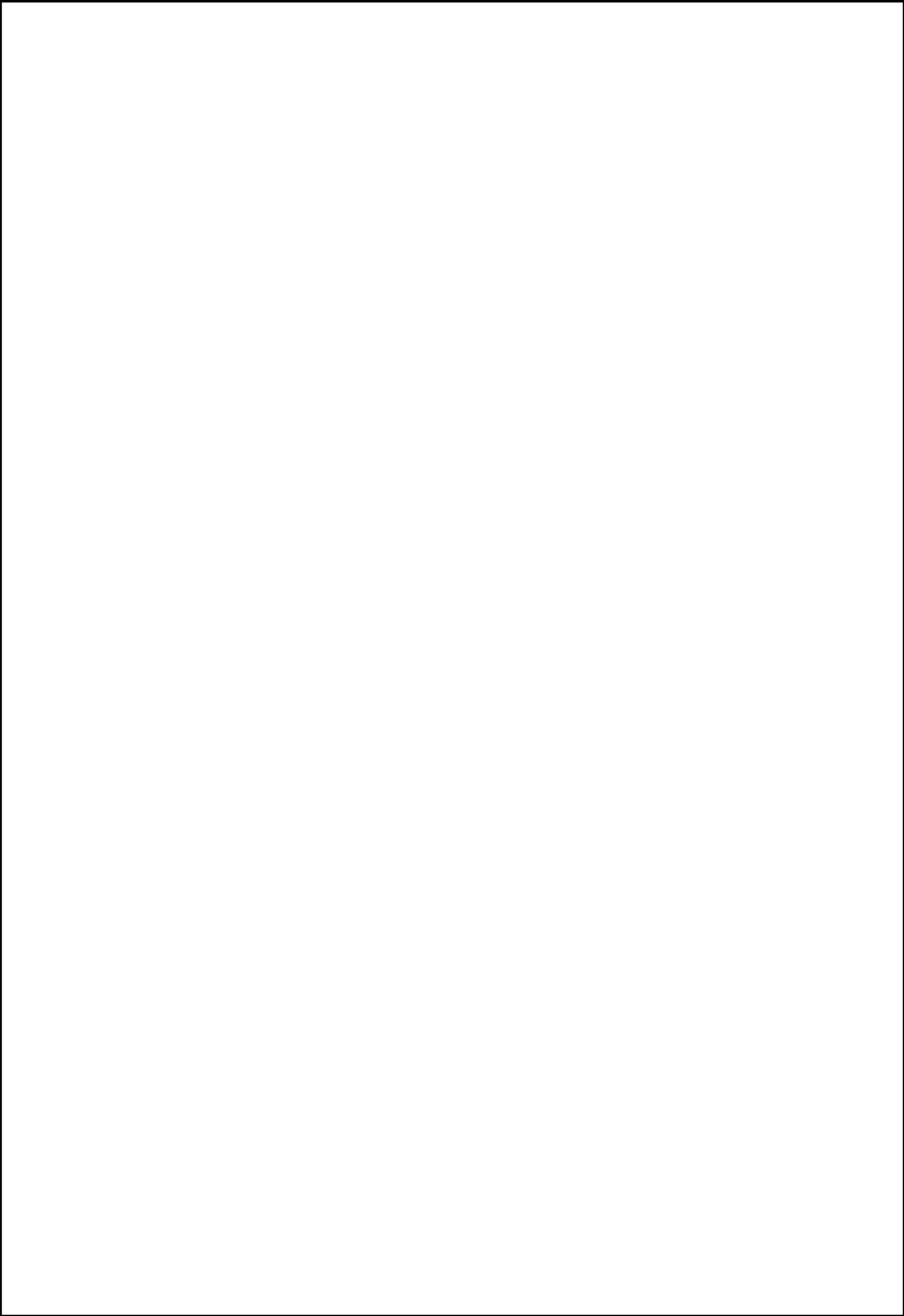
PRODUCTS SOLD PER COUNTRY

PAGE NO:

DATE:

AIM:

ALGORITHM:



SOURCE CODE:

```
C:\Users\STUDENT>cd..\..
```

```
C:\>mkdir C:\mapreduce\sales_mining
```

```
C:\>cd C:\mapreduce\sales_mining
```

```
C:\mapreduce\sales_mining>notepad sales.csv
```

```
C:\mapreduce\sales_mining>type sales.csv
```

```
C:\mapreduce\sales_mining>notepad sales_mapper.py
```

```
C:\mapreduce\sales_mining>notepad sales_reducer.py
```

```
C:\mapreduce\sales_mining>type sales.csv | python sales_mapper.py | sort | python  
sales_reducer.py
```

Sales.csv:

OrderID,Country,Product,Quantity

1001,India,Laptop,2

1002,USA,Phone,5

1003,India,Tablet,3

1004,USA,Laptop,1

1005,UK,Phone,4

1006,India,Phone,6

1007,UK,Tablet,2

1008,USA,Tablet,3

1009,India,Laptop,1

1010,UK,Laptop,5

Sales_mapper.py:

```
import sys
```

```
first_line = True
```

```
for line in sys.stdin:
```

```
    if first_line:
```

```
        first_line = False
```

```
        continue
```

```
    parts = line.strip().split(',')
```

```
if len(parts) == 4:

    country = parts[1]

    quantity = parts[3]

    print(country + '\t' + quantity)
```

Sales_reducer.py:

```
import sys

current_country = None

total = 0

for line in sys.stdin:

    country, qty = line.strip().split('\t')

    qty = int(qty)

    if current_country == country:

        total += qty

    else:

        if current_country:

            print(current_country + '\t' + str(total))

            current_country = country

            total = qty

        if current_country:

            print(current_country + '\t' + str(total))
```

OUTPUT:

India	12
UK	11
USA	9

RESULT:

EX.NO: 07

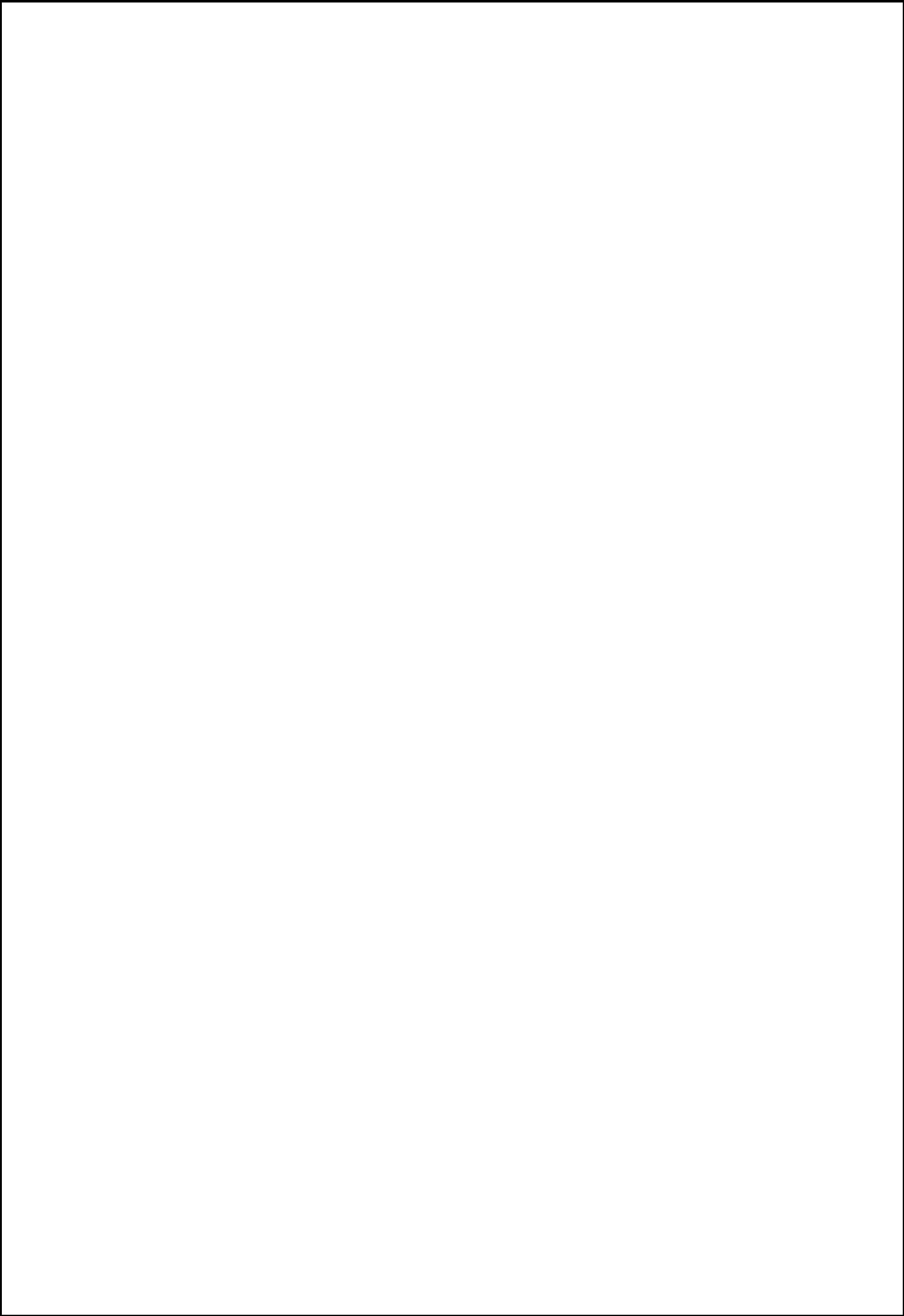
DATE:

**MAXIMUM ELECTRICAL CONSUMPTION
PER YEAR**

PAGE NO:

AIM:

ALGORITHM:



SOURCE CODE:

```
C:\Users\STUDENT>cd..\.
```

```
C:\>mkdir C:\mapreduce\electricity_mining
```

```
C:\>cd C:\mapreduce\electricity_mining
```

```
C:\mapreduce\electricity_mining>notepad electricity.csv
```

```
C:\mapreduce\electricity_mining>type electricity.csv
```

```
C:\mapreduce\electricity_mining>notepad elec_mapper.py
```

```
C:\mapreduce\electricity_mining>notepad elec_reducer.py
```

```
C:\mapreduce\electricity_mining>type electricity.csv | python elec_mapper.py | sort | python  
elec_reducer.py
```

electricity.csv

```
Year,Month,State,Consumption_MW
```

```
2019,January,TamilNadu,4500
```

```
2019,February,TamilNadu,4200
```

```
2019,March,TamilNadu,5100
```

```
2020,January,TamilNadu,5300
```

```
2020,February,TamilNadu,4900
```

```
2020,March,TamilNadu,5800
```

```
2021,January,TamilNadu,5500
```

```
2021,March,TamilNadu,6100
```

```
2022,January,TamilNadu,6300
```

```
2022,March,TamilNadu,6500
```

elec_mapper.py:

```
import sys
```

```
first_line = True
```

```
for line in sys.stdin:
```

```
    if first_line:
```

```
        first_line = False
```

```
        continue
```

```
    parts = line.strip().split(',')
```

```
if len(parts) == 4:

    year = parts[0]

    consumption = parts[3]

    print(year + '\t' + consumption)
```

elec_reducer.py:

```
import sys

current_year = None

max_val = float('-inf')

for line in sys.stdin:

    year, val = line.strip().split('\t')

    val = float(val)

    if current_year == year:

        if val > max_val:

            max_val = val

    else:

        if current_year:

            print(current_year + '\t' + str(max_val) + ' MW')

            current_year = year

            max_val = val

        if current_year:

            print(current_year + '\t' + str(max_val) + ' MW')
```

OUTPUT:

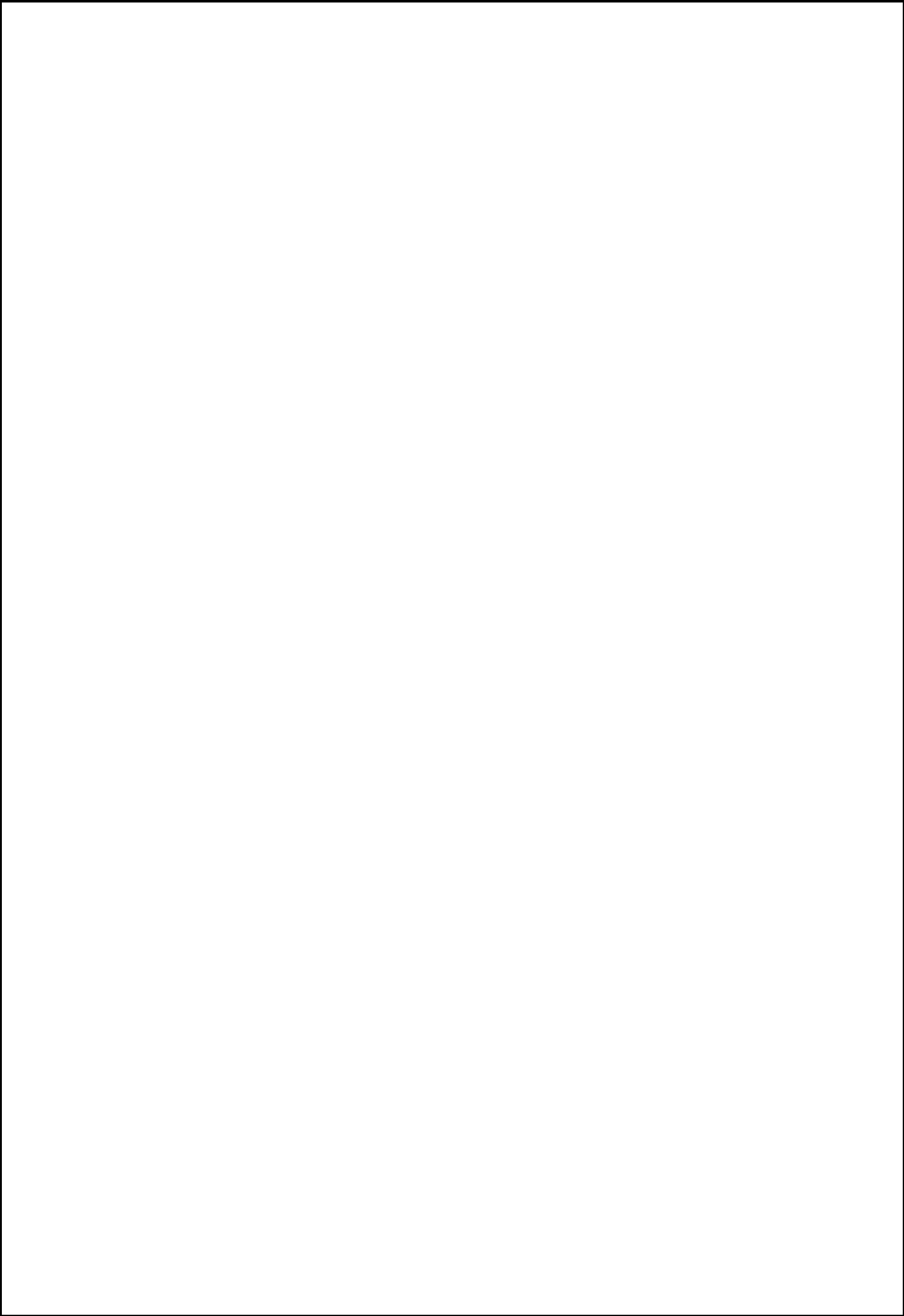
2019	5100.0 MW
2020	5800.0 MW
2021	6100.0 MW
2022	6500.0 MW

RESULT:

EX.NO: 08 DATE:	REAL - TIME LOG DATA ANALYSIS	PAGE NO:
----------------------------------	--------------------------------------	-----------------

AIM:

ALGORITHM:



SOURCE CODE:

```
C:\Users\STUDENT>cd..\.
```

```
C:\>mkdir C:\mapreduce\log_mining
```

```
C:\>cd C:\mapreduce\log_mining
```

```
C:\mapreduce\log_mining>notepad access.log
```

```
C:\mapreduce\log_mining>notepad log_mapper.py
```

```
C:\mapreduce\log_mining>notepad log_reducer.py
```

```
C:\mapreduce\log_mining>type access.log | python log_mapper.py | sort | python  
log_reducer.py
```

access.log:

```
2024-01-15 10:00:01 ERROR Database connection failed  
2024-01-15 10:00:05 INFO Request received from 192.168.1.1  
2024-01-15 10:01:10 ERROR Timeout exceeded for query  
2024-01-15 10:02:00 WARNING Memory usage at 85 percent  
2024-01-15 10:02:30 ERROR Disk full on drive D  
2024-01-15 10:03:00 INFO Scheduled task completed  
2024-01-15 10:03:45 WARNING CPU usage spike detected  
2024-01-15 10:04:00 ERROR Authentication failed  
2024-01-15 10:05:00 INFO Health check passed  
2024-01-15 10:06:00 ERROR Service unavailable
```

Log_mapper.py:

```
import sys
```

```
for line in sys.stdin:
```

```
    line = line.strip()
```

```
    if not line:
```

```
        continue
```

```
parts = line.split()

if len(parts) >= 3:

    log_level = parts[2]

    print(log_level + '\t1')
```

Log_reducer.py:

```
import sys

current_level = None

count = 0

for line in sys.stdin:

    level, val = line.strip().split('\t')

    if current_level == level:

        count += int(val)

    else:

        if current_level:

            print(current_level + '\t' + str(count))

            current_level = level

            count = int(val)

        if current_level:

            print(current_level + '\t' + str(count))
```

OUTPUT:

ERROR 5
INFO 3
WARNING 2

RESULT: